



Original article

What are the factors influencing recreational visits to national forest parks in China? Experiments using crowdsourced geospatial data

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ABSTRACT

Quantifying recreational visits to national forest parks (NFPs) and deciphering associating driving factors is significant for NFP protection, management, development, and planning; however, previous studies have primarily conducted questionnaire-based surveys or semi-structured individual interviews to gather limited data, while few have employed big data to effectively and efficiently estimate NFP visits and further explored correlated factors. This study used location-based service data from social media to quantify recreational NFP visits, and search for any associations with spatio-physical park factors selected based on previous constructive studies and NFP visitor perceptions of travel experiences posted on social media. A spatial error model with maximum likelihood estimation (SEM-MLE) was applied to investigate the significance. Sixty-eight (68) NFPs located in the Yangtze River Delta urban agglomeration (YRDUA) were selected as experimental cases. The results indicated that: (i) The number of NFP check-in visits was unevenly distributed across the YRDUA; (ii) NFP size and biodiversity (i.e., the number of vegetation and animal species) were not correlated with check-ins; (iii) Three NFP attributes (entrance fee, percent vegetation cover, and the presence of recreational water activities), one accessibility variable (potential visitors to NFP), and one transport variable (driving time to train station), had a significant influence on NFP check-ins, collectively explaining 49.69% of the observed variation. These findings can help NFP managers and planners understand the internal and external factors influencing recreational visits, thus more strategically informing nature-based tourism design and promotion campaigns.

1. Introduction

National forest parks (NFPs) constitute the highest level in China's forest park system with unique landscapes, valuable cultural heritage, and a favorable eco-environment (Chen et al., 2019). NFPs play an important role in protecting forest resources, providing a wide array of ecosystem services, such as improving air quality (Baumgardner et al., 2012), mitigating urban heat island effects (Ren et al., 2018), purifying water (de Mello et al., 2018), and reducing noise (Yuan et al., 2019).

Notably, a number of the cultural ecosystem services provided bolster nature-based (i.e., forest-based) tourism. Recreational visits to NFPs provide urban residents with opportunities to benefit from natural exposure, encouraging outdoor physical activity (e.g., walking, climbing, and cycling), providing recreational and esthetic enjoyment, relieving stress, and thus promoting physiological and psychological health, as well as a spiritual sense of well-being (Bielinis et al., 2019; Huang, 2014; Liu et al., 2019). NFPs help promote natural science knowledge and regional economic development (Chen and Nakama,

2013).

Rapidly urbanizing lifestyles over recent decades have resulted in increasingly fewer opportunities to interact with nature (Hartig et al., 2014); therefore, NFPs have become a leading destination for eco-based tourists. Accordingly, quantifying recreational NFP visits and identifying correlated driving factors influencing their recreational use is fundamental for managers and planners when designing budgeting, facility and amenity allocation, estimating demand, assessing tourism impact, tourism carrying capacity, management alternatives, and conservation of forest resources (Barros et al., 2019; Cessford and Muhar, 2003; Eagles and McCool, 2002).

Currently, recreational visit data are often spatiotemporally restricted, as implementing field techniques such as visitor surveys and on-site counts is both costly and time-intensive (Heberling and Templeton, 2009). Recent advances in social media and network analysis provide opportunities to explore recreational visits and visitor behavior based on crowdsourced geospatial data, providing information on user preferences, notable events, and real-time sharing of perceptions (Liang

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and Zhang, 2021; Sessions et al., 2016; Walden-Schreiner et al., 2018a; Wood et al., 2013). Specifically, social media enables visitors to share knowledge and experiences regarding travel attributes (e.g., entrance fees, transport, scenery, and other attractions) through photographs, emoticons, and written descriptions on their online profiles or travel blogs (Baym and Boyd, 2012; Filieri et al., 2015). Moreover, many social media platforms (e.g., Twitter, Facebook, Flickr, and Sina Weibo) support location-based services (LBS; e.g., check-in options). The intersection of social media and LBS has led to the development of location-based social networks or location-sharing services, allowing visitors to geotag their locations or content (Ruths and Pfeffer, 2014; Schwartz and Halegoua, 2015).

Many studies have applied LBS data from social media platforms, finding significant positive correlations between official visitor statistics and check-ins (Li et al., 2020; Sessions et al., 2016; Tenkanen et al., 2017; Walden-Schreiner et al., 2018b; Wood et al., 2013). Although several NFPs in China have visitor monitoring programs and publish visitor counts every few years, comprehensive and representative data are largely time and cost-prohibitive for NFPs; thus, crowdsourced geospatial big data can be reliably leveraged to estimate NFP visits and understand recreational use patterns.

With respect to the factors affecting NFPs' recreational use, previous studies have revealed the significance of numerous potential variables (see Section 2.3); however, their relative importance to forest park visits appears mixed. For example, Koo et al. (2013) investigated urban residents' preference for various forest park attributes in South Korea based on face-to-face interviews, revealing that biodiversity was the most influential factor when selecting urban forest park recreation. A systematic review by Chen and Nakama (2013) suggested that forest park's fresh air was strongly promoted to attract visitors, and the scenic landscape was another major motivating factor affecting visits. Aikoh et al. (2012) revealed that many considered social and meteorological factors when visiting a suburban forest park in Japan, with days of the week (e.g., Sundays and holidays) as the most influential factor, followed by weather conditions. Similarly, Šodková et al. (2020) suggested that the main drivers for visiting forests in the Czech Republic, according to a national survey, were for walking or outdoor enjoyment, especially during holidays or weekends.

These inconsistencies may be attributed to the questionnaire-based surveys or semi-structured individual interviews over a single or several specific sites adopted by these studies. Additionally, the attributes of each sampled forest park and the socioeconomic differences between respondents may also create mixed results. Therefore, a more comprehensive approach, considering a wider range of samples and various objective factors should be implemented to rapidly and inexpensively enrich traditional data from surveys and official statistics; however, although a few recent studies have applied crowdsourced geospatial social media data to examine the association between urban park visits and objective affecting factors (Li et al., 2020; Lyu and Zhang, 2019; Zhang and Zhou, 2018), nearly none have specifically addressed the spatio-physical factors affecting NFP recreational visits. Notably, NFPs are distinct from urban parks because of their significant differences in location, function, recreational amenities, and services (Zhang et al., 2021b).

The objectives of this study were first to quantify NFP recreational visits (from 2012 to 2019) through geotagged social media data (i.e., Weibo check-in data) and examine the significance of potential spatio-physical factors affecting visits. Sixty-eight (68) NFPs located in the Yangtze River Delta urban agglomeration (YRDUA), the largest in China, were selected to address the following research questions:

- > Are there spatial disparities in NFP check-in visits and densities across the YRDUA?
- > What are the correlations between selected spatio-physical factors and total NFP check-in visits?

- > Which variables can best predict NFP check-in visits, and what are their implications for NFP management and planning?

2. Methods and data

2.1. Study area

Urban agglomerations (the spread of cities into larger assemblages) have become the main form of urbanization in China. The YRDUA study area is located in the downstream region of the Yangtze River, and comprises the most developed urban agglomeration and active economic development region in China (Zhang et al., 2021a). Further, the YRDUA is also one of the six largest urban agglomerations globally (Lin et al., 2020). According to the agglomeration's development plan, the area contains Shanghai and other important cities located in the provinces of Jiangsu, Zhejiang, and Anhui (Fig. 1). YRDUA covers an administrative area of 211,700 km², maintains a total population of 150 million residents, and a 2018 GDP of 17.85 trillion yuan (~ 20% of China's national economy). There is a total of 81 NFPs located in YRDUA according to the National Forestry and Grassland Administration in 2016, of which four are located in Shanghai, 18 in Jiangsu Province, 28 in Zhejiang, and 31 in Anhui. In this study, 68 NFPs, for which Weibo check-in records were available, were selected for analysis.

2.2. NFP check-in visits

To evaluate recreational visits to NFPs, and their potential affecting factors, evidence from crowdsourced geospatial big data was amassed. Although globally, Twitter and Facebook are the most popular social networking services, in China, Sina Weibo, a hybrid of Twitter and Facebook, is the largest and most influential social media platform. Compared with other Chinese social media apps, Weibo has a much higher degree of user participation (Li et al., 2020). According to the Weibo User Development Report, there were 511 million monthly active users and 224 million daily active users in December 2020. Weibo added location-based services (i.e., check-in functions) in May 2012, and since then, users have been able to upload their real-time locations. It now contains the largest available geotagged dataset in China. Weibo users can share their texts and photographs with location information on the platform (check-ins), as well as interact and communicate with netizens. Here, Weibo check-in data was applied to represent the number of NFP recreational visits, as previous research has confirmed that it is an effective indicator of visits to green spaces (Li et al., 2020; Zhang and Zhou, 2018). Specifically, the latitudes, longitudes, names, and check-in visitors were obtained through the "obtain location details" application program interface (API) using the Python programming language. In total, 316,233 geotagged check-in visits across 68 NFPs were retrieved during the 7-year period from May 2012 to August 2019. Note that these check-ins do not directly reflect the real visitation. Not everyone visiting NFPs has a Weibo account or posts a visit with location information. Unfortunately, we do not have access to real NFP visitation data to verify its relationship with Weibo check-ins. Nevertheless, an empirical study by Shen et al. (2017) focusing on 87 urban parks in Shanghai, China, showed a significant correlation between Weibo check-ins and official visitor statistics. Similarly, a recent study by Li et al. (2020) also indicated that the correlation between the Weibo check-in data of urban parks and the official visitor statistics was significantly correlated, and the coefficient ranges between 0.5 and 0.8. Therefore, social media check-in data can be reliably used to estimate the NFP visit intensity (Fisher et al., 2018; Sessions et al., 2016).

2.3. Potential variables affecting NFP check-in visits

Two steps were conducted to determine the potential variables affecting NFP check-in visits. First, a range of valid potential impact predictors affecting recreational visits to NFPs or urban parks were



Fig. 1. The spatial distribution of the 68 National Forest Parks (NFPs; green circles) located in the Yangtze River Delta urban agglomeration (YRDUA), China.

derived from previous studies. These variables were divided into internal and external factors, with the former primarily referring to park attributes, such as size (Zhang and Zhou, 2018), vegetation quality (Wang et al., 2017), biodiversity (Koo et al., 2013), recreational facilities (Li et al., 2020), and entrance fees (Chen and Qi, 2018). The latter primarily include accessibility (Shen et al., 2019), transport (Hamstead et al., 2018), surrounding environments (Lyu and Zhang, 2019), location (Rossi et al., 2015), and weather (Aikoh et al., 2012).

Second, visitors' perceptions of their travel experiences can also reflect potential influencing factors to some degree (Oku and Fukamachi, 2006; Zhang and Chan, 2016). Text-mining methods examined the occurrence of keywords and semantic analysis, as evaluated through the ROST CM6 platform, to analyze visitor perceptions of travel experiences posted on social media. Data was acquired from short travel blogs (i.e., online reviews) from Ctrip, the largest travel website in China, with a 2014 market share of 36% (<http://www.ctrip.com>), for the text-mining analysis (Qi et al., 2017). The blog contents and associated comments represent targeted descriptions of each destination's attributes, potentially revealing visitor behavior, preference, and movement. A Python program was developed to collect the content data, acquiring 48,980 valid online reviews and travel blogs relating to the visitation perception of NFPs located in YRDUA. According to keyword occurrence (Table 1) and semantic analysis (Fig. 2), "air quality", "entrance fee", "scenery", "recreation", "transport", and "environment"

Table 1

Top 20 high-frequency keywords from visitors' online comments on Ctrip.

Number	Keywords	Frequency	Number	Keywords	Frequency
1	Air quality	5843	11	Children	1132
2	Entrance fee	2913	12	Interesting	1125
3	Climbing	2307	13	Happy	1116
4	Environment	2195	14	Beautiful	1089
5	Recreation	1759	15	Price	1077
6	Fresh	1527	16	Weather	914
7	Waterfall	1360	17	Amenities	810
8	Oxygen bar	1327	18	Transport	755
9	Mountain	1231	19	Bicycle	733
10	Barbecue	1155	20	Cableway	732

were the most important topics for tourists. Ultimately, 11 predictor variables were selected across four themes: NFP attributes, accessibility, transport, and location (described in more detail below; Table 2). It was hypothesized that these spatio-physical variables were significantly correlated with NFP check-in visits.

2.3.1. NFP attributes

The NFP attributes incorporated the percent vegetation cover, presence of water activities, entrance fees, NFP size, and biodiversity. Vegetation cover is a fundamental indicator for assessing NFP quality,

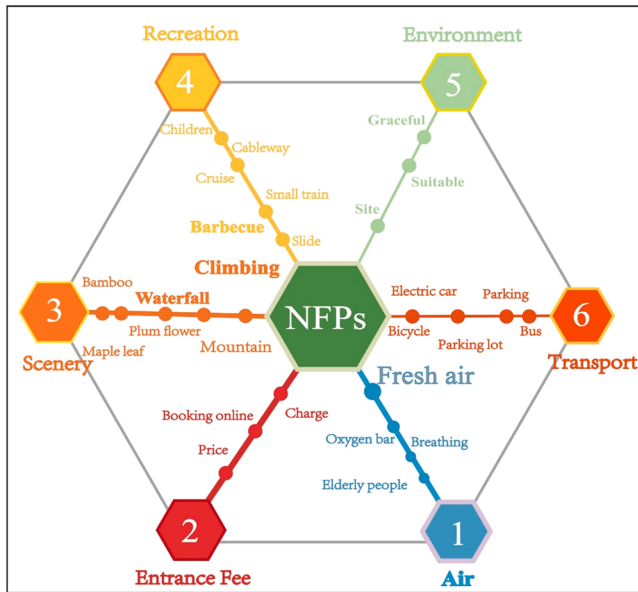


Fig. 2. The semantic network of retrieved keywords using visitors' online comments from Ctrip.

Table 2

Descriptions and data sources of selected variables.

Categories	Variables	Descriptions	Methods and data sources
NFPs attributes	Size	NFP size	Statistical yearbook of local authorities
	Fee	The price of entrance fee	Ctrip.com
	Pre_Water	Presence of water activities	Google Earth and Ctrip tour descriptions
	Veg_Rate	Vegetation cover	Statistical data from the national forestry authorities or NFP websites
	N_plant species ^a	Number of plant species	Statistical data from the national forestry authorities or NFP websites
	N_animal species ^a	Number of animal species	Statistical data from the national forestry authorities or NFP websites
Accessibility	Pro_Megacities	Proximity to (the urban center of) megacities ^b	Transportation database (i. e., road network database) from Bigemap ^c , including road network level (e.g., highways, major roads, secondary roads, and branches), road junctions, and traffic signal positions. The population database was collected from the sixth national population census.
	Pot_Visitors	Potential accessible visitors in YRDUA	Retrieved from Baidu Map API
Transport	N_Stops	Bus stop no. ≤ 1 km of the NFP entrances	Retrieved from Baidu Map API
	T_Train station	Driving time to the nearest train station	Retrieved from Baidu Map API
Location	D_Urban center	The (network) distance to the urban center	Estimated by the OD Cost Matrix Analysis in ArcGIS v.10.5 with the abovementioned transportation database

^a The number of plant and animal species samples were 51 and 45, respectively, because some NFPs have not published specific official statistics on them.

^b The top 10 GDP megacities located in YRDUA include Shanghai, Suzhou, Hangzhou, Nanjing, Wuxi, Ningbo, Nantong, Hefei, Changzhou, and Wenzhou.

^c Bigemap is a high-definition map downloader, with a pixel accuracy of 0.25 m that is updated every 3–6 months in China.

where high coverage rates contribute to improved air quality for visitors to enjoy (She et al., 2017). These data can be obtained by referring to the statistics of the national forestry authorities or the NFP websites. The presence of hydrological landscapes or water activities (e.g., waterfalls, canoeing, kayaking, rafting, etc.) is another crucial variable influencing visitation to NFPs, due to their significant contributions to recreation and esthetic enjoyment (Zhang et al., 2021b). We first used Google Earth to identify the presence of water features initially, and then checked the NFP tour descriptions on Ctrip to determine the presence of water activities. The price of entrance fees available on the NFP website or Ctrip can roughly reflect the quality and popularity of NFPs (Cohen et al., 2010; Zhang and Zhou, 2018). Moreover, they were also one of the most often mentioned topics based on the text-mining analysis. NFP size is considered because larger parks may provide more recreational services and facilities. We obtained the NFP size database from the statistical yearbook of local authorities or NFP websites. Lastly, biodiversity was considered to explore the impact of plant and animal species abundance on NFP recreational visits. The number of plant and animal species freely available in the statistical yearbooks of the national forestry authorities or on the NFP websites was used as an approximate proxy for the NFP biodiversity. Notably, although the presence of mountains or climbing activities were potential variables influencing NFP recreational visits (as suggested by the high-frequency keywords from visitor perceptions posted on social media), they were excluded here, as the vast majority of NFPs ($n = 61$) in the YRDUA are located in such areas.

2.3.2. Accessibility

NFP accessibility is a crucial external factor affecting recreational visits and assessed here based on two parameters: proximity to megacities, and the number of potential visitors to NFP in YRDUA. For the former, the minimum network distance from the NFP to the urban center of each megacity was used, and the cumulative network distance was estimated to represent each NFP's proximity to the megacities. For the latter, the Huff model, a widely accepted approach for quantifying the probability of people's selection of a service site out of multiple available sites (Huff, 1963, 1964), was adopted to estimate potential visitors to the NFPs. The Huff model is essentially gravity-based, and considers the effects of distance decay on the interactions between the origin and destination. Specifically, it aims to evaluate people's choice of destination considering two basic factors: (1) distance to the destination, and (2) the quality/attractiveness of the available site. The Huff model is given by Eq. (1):

$$Prob_i = \frac{M_j d_{ij}^{-\beta}}{\sum_{j=1}^n M_j d_{ij}^{-\beta}} \quad (1)$$

where $Prob_i$ is the probability of visitors from the original location i (here referring to the geometric centroids of district i in the YRDUA) accessing NFP j ; d_{ij} is the travel time between i and j ; β is the distance impedance coefficient, which is usually set at 2 according to previous studies (Talen and Anselin, 1998); and M_j is the capacity/attractiveness of NFP j . Notably, M_j is set based on the scenic spot level of the NFP, ranging from A to 5A, where higher levels indicate better ecological resources, forest landscapes, cultural landscape qualities, and greater scientific value as evaluated by the Chinese National Tourism Administration. A 5A-level NFP scenic spot was set to five, while an A-level scenic spot or area with no rating was set to 1.

Furthermore, the potential visitors to each NFP in the YRDUA were estimated according to Eq. (2):

$$V_j = \sum_{i=1}^m Prob_i \times P \quad (2)$$

where V_j represents the number of potential visitors to NFP j , P represents the population of the original location i , and m is the number of subdistricts in YRDUA.

2.3.3. Transport

Transport was considered in terms of two parameters: the number of nearby bus stops ≤ 1 km from the NFP entrance, and the driving time (i. e., distance) from the NFP entrance to the nearest train station. Baidu Map Web API v2.0 service for developers to obtain points of interest (POIs) was used to obtain geospatial bus stop information in HTTP/HTTPS format using Python, and nearby bus stops were collected using a 1 km buffer in ArcGIS v.10.5. In addition, route planning services were obtained based on Baidu maps to estimate the minimum driving times from the NFP entrance to the nearest train station.

2.3.4. Location

NFP location was evaluated in terms of the minimum network distance from the major NFP entrance to the urban center using the *OD cost matrix* analysis in ArcGIS.

2.4. Statistical analysis

Spearman's correlation was used to examine the bivariate correlations between NFP check-in visits and their potential influencing factors. An ordinary least squares (OLS) model was subsequently employed as a conventional regression method to analyze the impacts and relative importance of these factors on check-in visits. The variance inflation factor (VIF) for each predictor variable was calculated, and the mean VIF values were used to test for multicollinearity among the variables.

Additionally, according to Tobler's first geographic law, the closer the spatial distance is between objects (i.e., NFPs), the greater the correlation between their attributes' values. Given the likelihood of spatial autocorrelation of NFP check-in visits, the global Moran's index (I) was calculated to assess the relationships between the dependent variables in the initial OLS model, and returned significant results (Moran's $I = 0.206$, $p < 0.01$). Therefore, spatial regression analysis was

employed to further identify the most influential variables associated with check-in visits. To this end, the Lagrange multiplier test statistics suggested that the spatial error model (SEM) was the most appropriate in the present study (Anselin, 2005). The SEM model was calculated according to Eq. (3):

$$Y = X\beta + \rho W_e + \mu \quad (3)$$

where Y is the dependent variable representing the NFP check-in visits, X is the matrix of potential independent variables, β is the slope vector reflecting the impact of the explanatory variable on NFP check-in visits, ρ is the spatial autoregressive coefficient of the error term, W_e is the (Euclidean) distance-based spatial weight matrix, and μ is a vector of the error terms.

Response variables were transformed, as the NFP check-in data were non-normally distributed. Maximum likelihood estimation (MLE) was used to solve for the best-fit model (Mardia and Marshall, 1984). Regarding model verification and accuracy, Akaike Information Criterion (AIC) and log-likelihood statistics (LogL) were used to compare the results of OLS and SEM-MLE (Anselin, 2005) using the GeoDaSpace v.1.2 software platform (Anselin and Rey, 2014).

3. Results

3.1. Spatial disparities in NFP check-in visits and densities

A total of 316,233 check-in visits to NFPs located in YRDUA were reported on Sina Weibo from 2012 to 2019, averaging (4650 visits per NFP, ranging from 26 to 35,358). There were few NFPs ($n = 5$) with < 100 visits, with the vast majority maintaining 100–20,000 check-in visits (79.4%), of which 33.8% had 100–1000 visits, 26.5% had 1000–5000 visits, and 19.1% had 5000–10,000 visits. Further, 13.2% of the NFPs received $> 10,000$ check-in visits, of which 8.8% were between

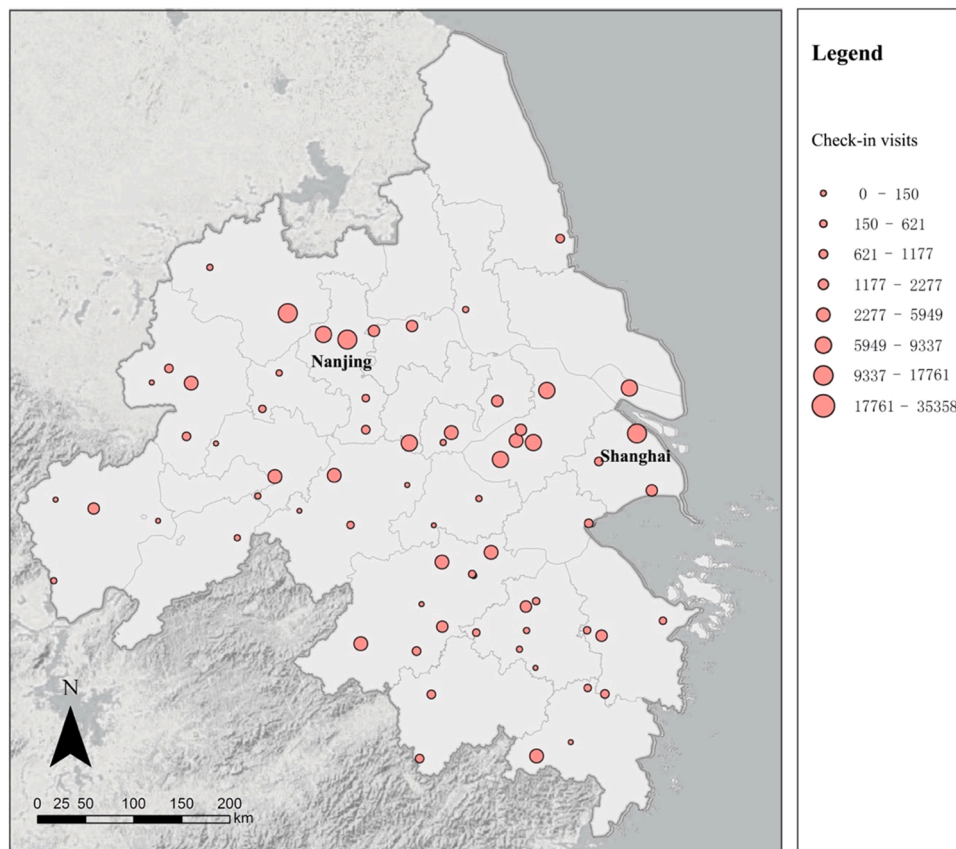


Fig. 3. The spatial distribution of NFP check-ins in the YRDUA.

10,000 and 20,000 visits, and 4.4% received > 20,000 visits.

The spatial distribution of NFP visits varied greatly (Fig. 3); generally, those located in Shanghai and Jiangsu Province had more check-in visits, while those located in the provinces of Zhejiang and Anhui were visited less frequently. The NFP with the most check-in visits was Zijin Mountain National Forest Park, located in Nanjing City, Jiangsu Province. It is the first large-scale urban forest park in China to be rated as an NFP. Of the top 10 most-visited NFPs, only the Langya Mountain NFP is not located in Shanghai or Jiangsu Province. Comparing the number of NFP check-in visits across different metropolitan areas, it was found that NFPs with the highest check-in visits were mostly concentrated in the Shanghai metropolitan area, followed by Nanjing, Suzhou-Wuxi-Changzhou, and the Hangzhou metropolitan areas. In contrast, the NFP check-in visits in the Hefei and Ningbo metropolitan areas were relatively low.

3.2. Correlations between selected variables and NFP check-in visits

Table 3 presents the results of Spearman's correlations between the NFP check-in visits and the selected predictor variables, where a higher absolute value of the standardized coefficient indicates a stronger relationship. The results suggested that the presence of water activities, vegetation cover, the potential visitors in the YRDUA, and the number of bus stops had significantly positive relationships with the NFP check-in visits. Conversely, the driving time to the nearest train station, proximity to megacities, and the distance to the urban center were negatively correlated with the NFP check-in visits (Fig. 4). It was further revealed that NFP biodiversity was insignificantly correlated with check-in visits ($p > 0.05$ for both plant and animal species), as were NFP size and entrance fees.

3.3. Results of OLS and SEM-MLE model

Table 4 presents the results of the OLS and SEM-MLE models. Notably, the VIF values of all independent predictor variables were < 7.5, suggesting weak multicollinearity among them. The results of the OLS model (Model 1) showed that four variables, including the presence of water activities, vegetation cover, potential visitors, and driving time to train station significantly influenced the number of NFP check-in visits (with only the latter maintaining a negative correlation). NFP size, location, proximity to megacities, and the number of bus stops held insignificant levels of influence on check-in visits. Overall, the adjusted R^2 of Model 1 was 0.4314, suggesting that approximately 43.14% of the variation in NFP check-in visits could be explained by this model.

With regard to the SEM-MLE model (Model 2), four variables had a positive influence on NFP check-in visits: entrance fee, vegetation cover, presence of water activities, and potential visitors. Driving time to train stations similarly showed a negative correlation with NFP check-in visits. The pseudo R^2 value of Model 2 was 0.4969, notably more capable of explaining the observed variance than Model 1. Moreover, both the AIC and LogL values indicated that the SEM-MLE model was superior.

Table 3
Correlations between check-in visits and the selected variables.

NFP	Size	Entrance Fee	Pre_Water	Veg_Rate
Check-ins	-0.153	0.155	0.470**	0.381**
	N_Plant species	N_Animal species	Pro_Megacities	Pot_Visitors
Check-ins	-0.54	-0.093	-0.265*	0.312**
	N_Stops	T_Train station	D_Urban center	
Check-ins	0.361**	-0.399**	-0.276*	

Note: Correlations are significant at the 0.05* and 0.01** level.

4. Discussion

4.1. Importance of NFP attribute factors in affecting check-in visits

Eleven (11) predictor variables of NFP attributes, accessibility, transport, and location were selected for correlations with check-in visits. The results of the SEM-MLE model suggested the significance of five variables: three NFP attribute variables (entrance fee, vegetation cover, and the presence of water activities), one accessibility variable (potential visitors), and one transport variable (driving time to train station). The relative importance of each significant variable is discussed below.

Although higher usage fees may theoretically decrease park accessibility to certain segments of tourists (Chase et al., 1998; Lal et al., 2017), the price of NFP entrance fees were positively associated with check-in visits in our model. This observation is in line with Ghermandi and Fichtman (2015) who found that natural wetlands with entrance fees had higher visitor numbers. Similarly, an empirical study focused on city parks also reported that those with entrance fees received more visitors (Zhang and Zhou, 2018). One possibility is that the NFP with higher entrance fees may have more impressive landscapes, or better recreational and educational facilities, thus increasing its desirability to eco-based tourists.

The significance of vegetation cover and the presence of water activities on NFP check-in visits are supported by Kang et al. (2019) who suggested that visitors attach the greatest importance to the natural environment of NFPs. Moreover, according to Chen et al. (2018), air quality is one of the most attractive features for attracting NFP visitors, and higher vegetation coverage can more efficiently purify and increase the negative oxygen ion concentrations in the air (De Carvalho and Szlafsztein, 2019; Qin et al., 2019). Additionally, the documented significance of water activities on NFP check-in visits aligned with a Western empirical study by Hamstead et al. (2018) who adopted geo-tagged Flickr and Twitter data to estimate the variation of use across New York City's 2143 diverse parks, finding a positive correlation with the presence of water bodies. This observation may be related to people's higher preferences for recreational and physical activities in wilderness containing water-related landscapes or activities (Dramstad et al., 2006; Kaltenborn and Bjerke, 2002). Conversely, the remaining two significant NFP attribute variables, biodiversity and NFP size, failed to support the hypothesis. Specifically, NFP plant and animal biodiversity was not significantly correlated with the number of check-in visits. This finding is contrary to that of Koo et al. (2013) who conducted face-to-face interviews to examine the urban residents' preferences of urban forest recreational services, concluding that biodiversity was the most influential indicator. These inconsistencies may be explained by the fact that the biodiversity variable only considered the number of vegetation and animal species, in addition to the lack of official statistics on species diversity in about 25% of NFPs in our study areas.

Regarding the NFP size, many studies have suggested positive relationships between green space area (primarily in urban parks) and recreational visits, as larger parks tend to have more trees, walking paths, water features, and other recreational facilities (Giles-Corti et al., 2005; Zhang et al., 2021b; Zhang and Zhou, 2018). For example, Li et al. (2020) evaluated visitor number through an analysis of Sina Weibo check-in data for 13,759 parks located in all 287 cities across China, finding that larger park size (i.e., larger service areas) had the greatest positive impact on urban park check-in visits; however, the findings of the present analysis revealed no such correlation. It was speculated that a size threshold existed here, where within this area, increasing area leads to more NFP visits; however, once the threshold is exceeded, no significant correlation between visits and park size remained. In other words, visitors could enjoy various types of cultural ecosystem services and recreational facilities afforded by green spaces once the parks are sufficiently large. Comprising the highest level of forest parks in China, NFP areas are typically large, with sufficient environmental amenities

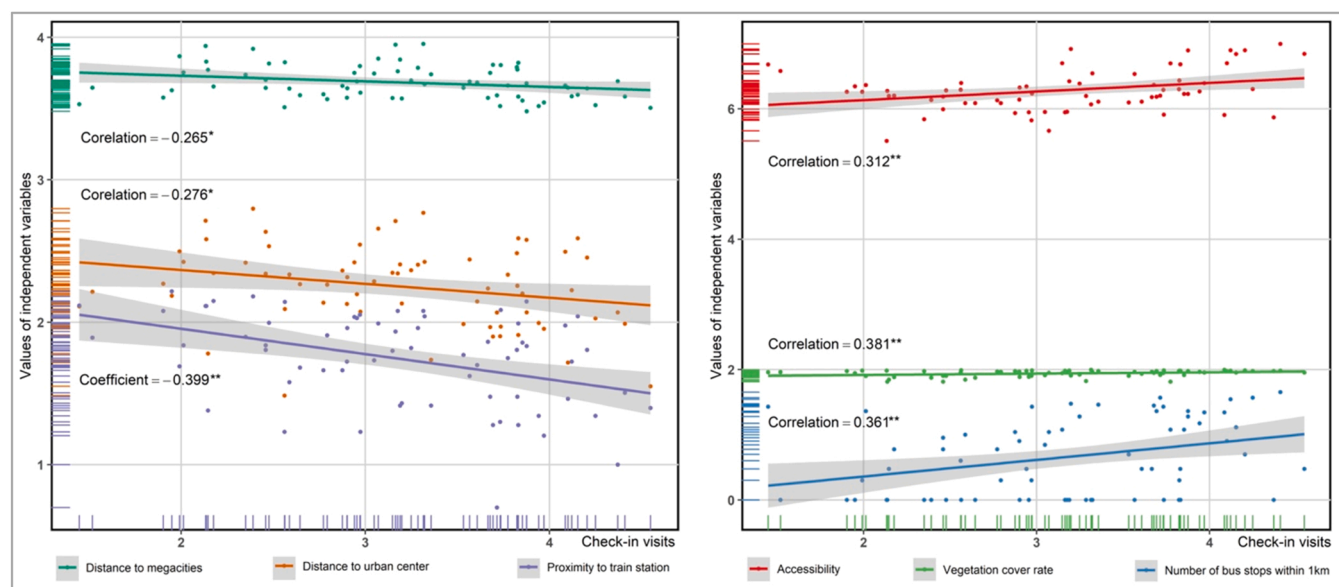


Fig. 4. The regression model relationships between check-in visits and the revealed significant predictors (one-factor): (a) negative correlations, (b) positive correlations. Notably, correlations between check-in visits and water activities were excluded as it was a categorical variable of presence (1) or absence (0).

Table 4
Results of OLS and SEM-MLE model.

Variables	Model 1: OLS			Model 2: SEM-MLE		
	Coefficient	Std. Error	p	Coefficient	Std. Error	p
Size	-0.053	0.158	0.736	-0.011	0.140	0.937
Entrance Fee	0.145	0.086	0.088	0.206**	0.071	0.004
Pre_Water	1.366**	0.378	0.000	1.429**	0.311	0.000
Veg_Rate	4.351*	1.665	0.011	4.560**	1.330	0.000
Pro_Megacities	0.259	0.859	0.764	0.382	0.707	0.589
Pot_Visitors	0.437*	0.251	0.087	0.494*	0.206	0.016
N_Stops	0.111	0.155	0.479	0.148	0.135	0.277
T_Train station	-0.869**	0.297	0.005	-1.142**	0.226	0.000
D_Urban center	0.162	0.369	0.663	0.489	0.283	0.084
Constant	1.411	9.074	0.877	-1.587	7.693	0.837
Adjusted/ Pseudo R ²	0.4314			0.4969		
AIC	239.179			234.406		
LogL	-109.590			-107.203		

Note: Correlations are significant at the 0.05* and 0.01** level.

and recreational activities; therefore, the size factor among NFPs has lessened impact on the number of NFP check-in visits.

4.2. Implications for NFP management and planning

The present study's findings are significant for NFP management and planning. First, some low-cost or free NFPs could be enhanced by charging appropriate entrance fees to help construct, or improve the maintenance of their recreational and educational facilities (Ghermandi and Fichtman, 2015). For example, NFP managers and planners can develop nature science education resources through this increased revenue to attract more nature-based tourists, and NFP brochures could be designed containing information on the value of local vegetation and biodiversity. Second, NFP natural environment, such as vegetation cover and the presence of water bodies for correlated activities, were the most influential factors affecting NFP check-in visits. Accordingly, forest managers should carefully evaluate tradeoffs between protecting the natural environment, and the development of human-modified 'gray' facilities. Third, transportation facilities and services, such as the surrounding bus stops, parking lots, and in particular, the driving time to

train stations, should be optimized to encourage NFP recreational use. As a nature-based tourism destination, recreational visits to NFPs attract large numbers of distant visitors from nearby cities (Shen et al., 2019) that are more likely to take the train (e.g., high-speed rail) for short destination trips in China (Li et al., 2019; Yang and Li, 2019). Finally, our results suggested that the accessibility variable (i.e., the potential number of visitors) evaluated by the modified Huff model is significantly associated with NFP check-in visits. In other words, this model has certain validity and credibility in estimating the number of NFP visits, which provides an important indicator for forest managers and planners to design appropriate recreational amenities and facilities, while controlling the tourism carrying capacity.

4.3. Limitations and future studies

Several limitations were present in this research. First, although previous studies have confirmed the effectiveness and availability of crowdsourced geospatial big data in both Western and Eastern countries (Fisher et al., 2018; Sessions et al., 2016; Shen et al., 2017; Zhen et al., 2017), comprehensive and consistent NFP visitation data were unavailable to validate the precise relationships with the social media check-in data (Wilkins et al., 2021). The estimation of NFP recreational visits based on Sina Weibo check-in data may underestimate the total visits due to check-in data biases among Weibo users by age, gender, education, or other demographic characteristics. For example, according to the Weibo Data Center (2019), 89% of all Weibo users are between the ages of 18–40 years old, suggesting that the check-in data may ignore visits older populations, a crucial potential visiting group seeking health-related benefits. Accordingly, future studies could design on-site questionnaires to explore the actual patterns of NFP recreational use. Second, regarding the factors affecting NFP attributes, lower levels of attention were given to esthetic and cultural factors, although NFPs represent the highest level of forest parks in China, have both beautiful forests and cultural landscapes. In addition, NFP accessibility was evaluated via the estimated number of potential visitors based on the Huff model, potentially underestimating visits due to the "edge effect". People living outside of the study area in nearby cities are likely also attracted to the YRDUA NFPs, and future models should be proposed to evaluate NFP accessibility more comprehensively. Overall, the revealed affecting factors explained ~49.69% of the variation observed in NFP check-in visits, and additional factors should be included in future

research to further examine these associations.

5. Conclusions

NFPs, the highest tier of China's forest park system, provide a wider range of cultural ecosystem services, such as recreational, physical, and educational activities, and have thus become one of the leading destinations for nature-based tourism in the country. Quantifying NFP recreational visits and deciphering the most influential determining factors are significant for sustainable NFP development. This study employed crowdsourced geospatial Weibo LBS data as a novel indicator to quantify NFP check-in visits; further, the spatial-physical factors affecting NFP check-in visits were also explored. The largest urban agglomeration in China, the YRDUA, was selected as the study area, and the results suggested that five variables maintained a significant influence on NFP check-in visits: three NFP attribute variables, one accessibility variable, and one transport variable. Collectively, these five internal and external NFP variables explained ~ 49.69% of the observed variation in check-in visits based on an SEM-MLE model.

The findings here have significant implications for NFP management and planning. NFPs with free or low-cost entrance fees are encouraged to charge appropriate amounts to help fund improvements to their recreational and educational facilities, thus attracting more nature-based tourists. Moreover, the trade-offs between protection of the natural environment and development of human-modified 'gray' facilities should be addressed, as a balance between the two appeared optimal. Lastly, the transport facilities and accessibility of NFPs must be considered during developmental planning, as such factors appeared vital in predicting total visitors. Future studies should incorporate more potential driving variables, particularly NFP cultural factors (e.g., historical or cultural sites), and landscape esthetic factors (e.g., color richness), to more comprehensively decipher the variation in NFP recreational visits.

CRedit authorship contribution statement

Jinguang Zhang: Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing. **Yingyi Cheng:** Conceptualization, Writing – original draft, Writing – review & editing. **Yuheng Mao:** Writing – review & editing. **Weizhen Cai:** Writing – review & editing. **Bing Zhao:** Conceptualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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